Part III

Cognitive and Behavioral Dysfunction

11

Memory Dysfunction in Dementia

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Although once thought to be a simple concept, memory is now considered to be a collection of mental abilities that use different systems and components within the brain to retain information over time. Memory research that began with neuropsychological studies of patients with focal brain lesions and now includes newer methods such as positron emission tomography, functional MRI, and event-related potentials has provided the rationale for a more refined and improved classification system (Budson, 2009; Schacter et al., 2000). In this chapter a number of different memory systems and their relevance for dementia will be reviewed, including the important anatomical structures for each, and the major disorders that disrupt them (Tables 11.1 and 11.2). A greater understanding of these memory systems will help the student or researcher to learn more about dementia in addition to aiding clinicians in their diagnosis and treatment of the memory disorders of patients. This improved understanding will become increasingly important as new therapeutic interventions for memory disorders are developed.

A memory system is a way for the brain to process information that will be available for use at a later time (Schacter & Tulving, 1994). Some systems are associated with conscious awareness (explicit) and can be consciously recalled (declarative), whereas others are typically unconscious (implicit) and are instead expressed by a change in behavior (non-declarative) (Squire, 1992). Memory can also be categorized in other ways, such as the nature of the material to be remembered, verbal (Champod & Petrides, 2010; Wagner et al., 1998) versus visual (Brewer et al., 1998; Morita et al., 2010).

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Table 11.1 Selected memory systems

Memory system	*Examples*	*Awareness*	*Length of storage*	*Major anatomical structures*
Episodic memory	Remembering a short story, what you had for dinner last night, and what you did on your last birthday	Explicit Declarative	Minutes to years	Medial temporal lobe, anterior thalamic nucleus, mamillary body, fornix, prefrontal cortex
Semantic memory	Knowing who was the first US President, the color of a lion, and how a fork and comb are different	Explicit Declarative	Minutes to years	Inferior lateral temporal lobes
Procedural memory	Driving a standard transmission car, and learning the sequence of numbers on a touch-tone phone without trying	Implicit Non-declarative	Minutes to years	Basal ganglia, cerebellum, supplementary motor area
Working memory	Phonological: keeping a phone number "in your head" before dialing Spatial: Mentally following a route, or rotating an object in your mind	Explicit Declarative	Seconds to minutes; information actively rehearsed or manipulated	Phonological: prefrontal cortex, Broca's area, Wernike's area. Spatial: Prefrontal cortex, visual association areas

Table 11.2 Selective memory system disruptions in common clinical disorders

Disease	*Episodic memory*	*Semantic memory*	*Procedural memory*	*Working memory*
Alzheimer's disease	+++	++	–	++
Frontotemporal dementia	++	++	–	+++
Semantic dementia	+	+++	?	–
Lewy body dementia	++	?	?	++
Vascular dementia	+	+	+	++
Parkinson's disease	+	+	+++	++
Huntington's disease	+	+	+++	+++
Progressive supranuclear palsy	+	+	++	+++
Korsakoff syndrome	+++	–	–	+/–
Multiple sclerosis	+	+/–	?	++
Transient global amnesia	+++	+/–	–	–
Hypoxic-ischemic injury	++	–	–	+/–
Head trauma	+	+	+/–	++
Depression	+	+/–	++	+/–
Anxiety	+	–	–	+/–
Obsessive compulsive disorder	+	–	++	++
Attention deficit hyperactivity disorder			?	+

+++ = Early and severe impairment; ++ = Moderate impairment; + = Mild impairment; occasional impairment or impairment in some studies but not others; – = No significant impairment; ? = Unknown

Episodic Memory

Episodic memory refers to the explicit and declarative memory system used to remember a particular episode of one's life, such as going out to dinner with a friend. This memory system is dependent upon the medial temporal lobes (including the hippocampus), as episodic memory has been largely defined by what patients with medial temporal lobe lesions cannot remember relative to healthy individuals. Other critical structures in the episodic memory system (some of which are associated with a circuit described by Papez in 1937) include the basal forebrain with the medial septum and diagonal band of Broca, the retrosplenial cortex, the presubiculum, the fornix, mammillary bodies, the mammillothalamic tract, and the anterior nucleus of the thalamus (Mesulam, 2000) (Figure 11.1, plate section). A lesion in any one of these structures may cause the impairment that is characteristic of dysfunction of the episodic memory system.

Memory loss due to dysfunction of the episodic memory system generally follows a pattern known as Ribot's law, which states that events just prior to an ictus are most vulnerable to decay, whereas remote memories are more resistant. Thus, dysfunction of the episodic memory system typically causes greatest disruption in the ability to learn new information (anterograde amnesia), moderate disruption in the

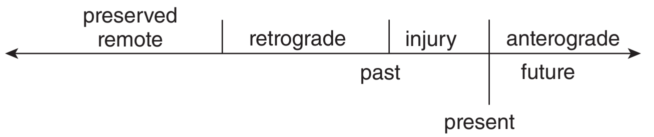

Figure 11.2 Ribot's law

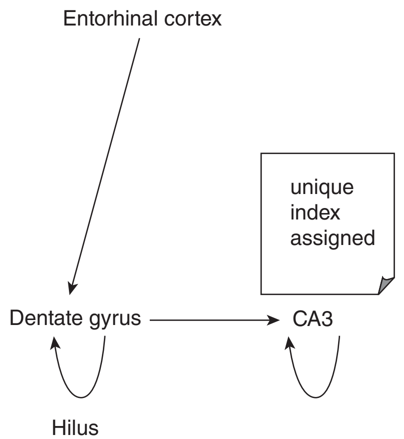

Figure 11.3 Schematic of encoding in the medial temporal lobe

ability to recall recently learned information (retrograde amnesia), and the ability to recall remotely learned information is generally intact (Ribot, 1881) (Figure 11.2).

The core of the episodic memory system is the medial temporal lobe and hippocampus. The medial temporal lobes are neuroanatomically complex structures with multiple regions and subregions including the parahippocampal gyrus, presubiculum, subiculum, and hippocampus proper, including its subregions.

Although we do not completely understand how the medial temporal lobes store and retrieve memories, our current understanding from cognitive neuroscience is as follows. An individual experiences an episode of their life, such as having lunch one afternoon. The cortically distributed patterns of neural activity representing the sights, sounds, smells, tastes, emotions, and thoughts during that episode are transferred first to the parahippocampal region and then to the hippocampus proper. After being transferred to the entorhinal cortex, the information is processed in the dentate gyrus, and then transferred to the CA3 region where it is further processed (Figure 11.3). It is in this CA3 region where the critically important hippocampal index is assigned, allowing the memory to be stored in a unique way so that it can later be recalled.

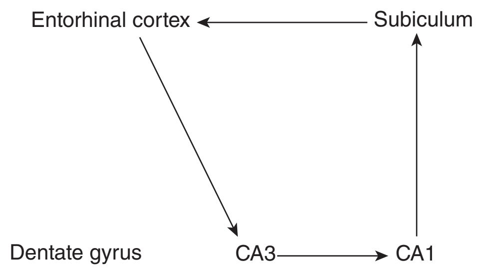

Figure 11.4 Schematic of retrieval in the medial temporal lobe

Typically memories are retrieved when a cue from the environment matches a part of the stored memory. Continuing our lunch example, years later the individual might now bite into a little cake that tastes remarkably like the one she had previously for desert during that particular lunch. This sensory cue is transferred from the cortex to the parahippocampal region and to the hippocampus. After the cue is transferred from the entorhinal cortex it now goes directly to the CA3 region where the original hippocampal index is retrieved (Figure 11.4). When found, the hippocampal index may be used to retrieve much of the original pattern of the neural activity representing the original episode stored in memory. This retrieved pattern of activity may then be transferred to the CA1 region, the subiculum, the entorhinal cortex, and then back out to the cortex – recreating all of sights, sounds, smells, tastes, emotions, and thoughts of the original memory episode.

The hippocampus remains critical for memory retrieval until a process known as *consolidation* occurs. Much research still needs to be done to better understand consolidation, but one thought is that once a memory is consolidated the distributed pattern of cortical neural activity is directly linked together, such that when a cue is encountered, the memory may be retrieved directly from cortical–cortical connections, without the need for the hippocampus. And although there are many details that need to be learned, there is much data that suggest that sleep is critical for consolidation to occur (Aly & Moscovitch, 2010; Diekelmann & Born, 2010; O'Neill et al., 2010; Stickgold, 2005, 2006).

In looking at the cognitive neuroscience model just presented of how memories are stored and retrieved, it is the CA3 region of the hippocampus that seems most critical – it is this region in which the hippocampal index is formed, and this region in which pattern matching of cues and memories occurs. Although we do not know exactly how the CA3 region is involved in these activities, a number of models using neural networks have been proposed. A simplified example of such a model is as follows. In this example we will show how an individual can find their car when they park in a parking garage over three successive days – and also why it is sometimes difficult for an individual to find their car. In this mythical parking garage there are two areas, Red and Blue, and two levels, first and second. Two weather

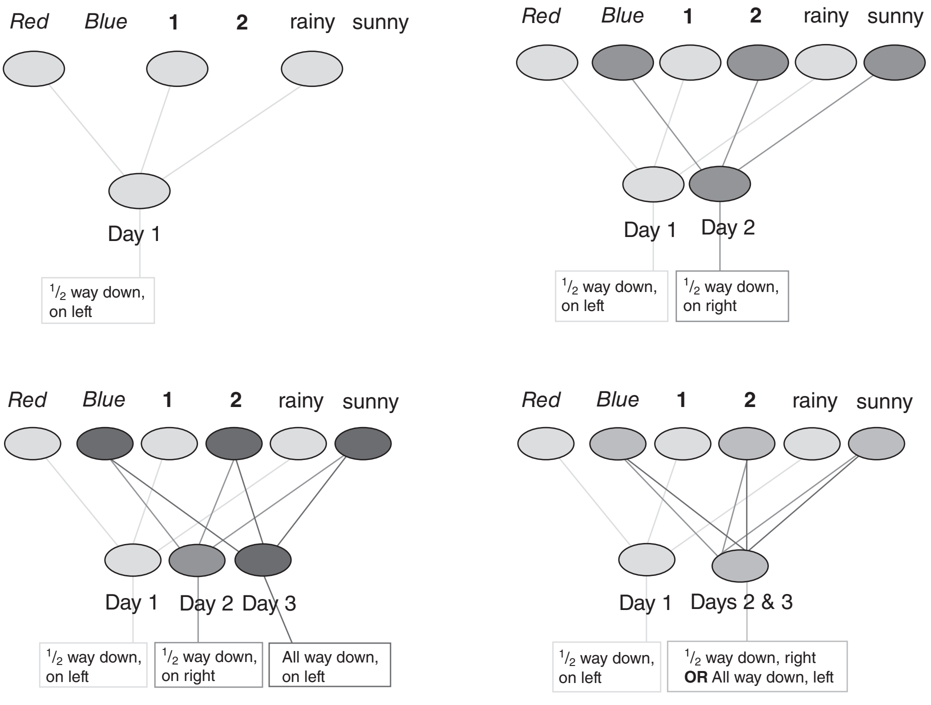

Figure 11.5 A neural network model. See text for details

states, rainy and sunny, are also shown to represent not only the weather but also other contextual details that may differ between one day and the next. On the first day the car is parked in the Red area, on the first floor, and it was a rainy day. From this distributed pattern of neural activity a hippocampal index can be formed that helps one remember that the car was parked in the Red area, on the first floor, and halfway down the aisle on the left (Figure 11.5, top left). On the second day the car is parked in the Blue area, on the second floor, and it was a sunny day. This distinct pattern of neural activity allows a unique hippocampal index to form that enables one to remember that the car was parked in the Blue area, on the second floor, and halfway down the aisle on the right (Figure 11.5, top right). On the third day the car is also parked in the Blue area, on the second floor, and it again was a sunny day (Figure 11.5, bottom left). Although one might wish that a hippocampal index will form to enable one to remember that the car was parked in the Blue area, on the second floor, and all the way down the aisle on the left – there is a problem. When there are completely overlapping patterns of neural activity a separate hippocampal index cannot form. Instead, there is a single hippocampal index which forms for both days two and three (Figure 11.5, bottom right). This hippocampal index is strengthened for the common aspects of the two memories: parking in the Blue area, on the second floor. But this index will also contain divergent aspects of the memory: halfway down the aisle on the right and all the way down the aisle on the left. Thus, on day three it will be easy to remember that the car is parked in the Blue area on the second floor, but it will be difficult to remember if it is parked halfway down the aisle on the right or all the way down the aisle on the left.

Over the last 15 years it has become increasingly clear that in addition to the medial temporal lobes and Papez's circuit, the frontal lobes are also important for episodic memory (Fletcher & Henson, 2001; Simons & Spiers, 2003). Whereas the medial temporal lobes are critical for the retention of information, the frontal lobes are important for the acquisition, registration, or encoding of information (Wagner et al., 1998); the retrieval of information without contextual and other cues (Manenti et al., 2010; Petrides, 2002); the recollection of the source of information (Johnson, Kounios et al., 1997); and the assessment of the temporal sequence and recency of events (Kopelman et al., 1997). One important reason why the frontal lobes are highly involved in episodic memory is that they enable the individual to focus their attention on the information to be remembered and to engage the medial temporal lobes. Also notable is that the left medial temporal and left frontal lobes are most active when a person is learning words (Wagner et al., 1998), and that the right medial temporal and right frontal lobes are most active when learning visual scenes (Brewer et al., 1998).

Dysfunction of the frontal lobes may cause a variety of memory problems, including distortions of episodic memory and false memories, such as when information becomes associated with the wrong context (Johnson, O'Connor et al., 1997) or incorrect specific details (Budson et al., 2002). Extreme memory distortions are often synonymous with confabulations, which occur when "memories" are created to be consistent with current information (Johnson, O'Connor et al., 1997), such as "remembering" that someone broke into the house and rearranged household items.

A clinically useful analogy can be used to help conceptualize the dysfunction in episodic memory that occurs due to damage to the medial temporal lobes (and Papez's circuit) versus damage to the frontal lobes (Budson & Price, 2002, 2005). The frontal lobes are analogous to the "file clerk" of the episodic memory system, the medial temporal lobes to the "recent memory file cabinet," and other cortical regions to the "remote memory file cabinet" (Table 11.3). Thus, if the frontal lobes are impaired, it is difficult – but not impossible – to get information in and out of storage. For example, getting information into storage may require stronger encoding, and getting information out of storage may require stronger cues from the environment. Additionally, when the frontal lobes are impaired the information

Table 11.3 A filing analogy of episodic memory

Brain structure	*Analogy*
Frontal lobes	File clerk
Medial temporal lobes	Recent memory files
Other cortical regions	Remote memory files

stored in memory may be distorted due to "improper filing" that leads to an inaccurate source, context, or sequence. If, on the other hand, the medial temporal lobes are impaired, it may be impossible for recent information to be stored. This will often lead to the patient asking for the same information again and again – perhaps 20 times in an hour. Older information that has been consolidated over months to years is likely stored in other cortical regions and will therefore be available for retrieval even when the medial temporal lobes or Papez's circuit are damaged. To illustrate this analogy we can compare the episodic memory dysfunction attributable to Alzheimer's disease versus depression. In general, patients with Alzheimer's disease have a dysfunctional "recent memory file cabinet," whereas patients with depression have a dysfunctional "file clerk."

Recent evidence has suggested that the parietal lobes also play an important role in episodic memory. In fact, in functional imaging studies the parietal lobes are more frequently associated with successful memory retrieval than either the frontal or medial temporal lobes (Simons et al., 2008). Despite this ubiquitous activation, the precise role of the parietal lobes in memory is not clear. Theories include attention to memory in both top-down and bottom-up roles (Cabeza et al., 2008), working memory's contribution to episodic memory (Berryhill & Olson, 2008), its role in retrieval and recollection (Davidson et al., 2008; Vilberg & Rugg, 2008, 2009), and the subjective memorial experience (Ally et al., 2008; Simons et al., 2010).

Evaluating episodic memory

When a disorder of episodic memory is suspected due to inability to remember recent information and experiences accurately, further evaluation is warranted. A detailed history of the memory dysfunction should be taken, with particular emphasis on the time course of the memory disorder. Speaking with a caregiver or other informant is usually critical, since the patient with memory dysfunction will invariably not remember important aspects of the history. A history of other cognitive deficits (such as deficits in attention, language, visuospatial, and executive function) should be obtained. Medical and neurological examinations should be performed, searching for signs of systemic illness, focal neurological injury, and neurodegenerative disorders.

The history, examination, and cognitive testing will suggest a differential diagnosis, which in turn will determine which laboratory and imaging studies are indicated. Brief cognitive testing may be performed by asking the patient to remember several words or a short story, or by using tools such as the Mini-Mental State Examination (MMSE; Folstein et al., 1975) – but beware of ceiling and floor effects (Franco-Marina et al., 2010), the Blessed Dementia Scale (Blessed et al., 1968), the Three Words–Three Shapes memory test (Mesulam, 2000), the word list memory test of the Consortium to Establish a Registry for Alzheimer's Disease (Welsh et al., 1992), the Drilled Word Span Test (Mesulam, 2000), the Seven-

Minute Screen (Solomon et al., 1998), and the Montreal Cognitive Assessment (www.mocatest.org). To help distinguish episodic memory dysfunction attributable to impairment of the frontal lobes versus impairment of the medial temporal lobes, difficulties in the encoding and retrieval of information should be contrasted with a primary failure of storage. When information cannot be remembered even when multiple rehearsals have maximized encoding, and retrieval demands have been minimized with the use of a multiple-choice recognition test, a primary failure of storage is present. In complex cases, a formal neuropsychological evaluation should be obtained.

Treatment depends upon the specific disorder identified. Cholinesterase inhibitors have been approved by the Food and Drug Administration (FDA) to treat Alzheimer's disease (Darreh-Shori & Jelic, 2010; Winblad et al. 2001) and Parkinson's disease dementia (Darreh-Shori & Jelic, 2010; Emre et al., 2004; Press, 2004); these medications have also been used to treat vascular dementia (Moretti et al., 2002) and dementia with Lewy bodies (McKeith et al., 2000). Memantine has been approved to treat Alzheimer's disease, with or without concomitant treatment with cholinesterase inhibitors (Tariot et al., 2004).

Episodic memory dysfunction in dementia

Episodic memory loss in degenerative diseases, such as Alzheimer's disease, dementia with Lewy bodies, and frontotemporal dementia, begins insidiously and progresses gradually (Solomon & Budson, 2003). Episodic memory in disorders that affect multiple brain regions, such as multiple sclerosis and vascular dementia, generally deteriorates in a stepwise manner. Some disorders of memory can have a more complicated and variable time course, including memory dysfunction attributable to tumors, hypoglycemia, medications, Korsakoff's syndrome, and traumatic brain injury. The cognitive profile of each disorder varies with the brain regions that are pathologically involved.

Alzheimer's disease

Alzheimer's disease pathology has a predilection for a number of particular regions of the brain. These regions include the hippocampus and amygdala, and the parietal, temporal, and frontal lobes. Alzheimer's pathology also affects subcortical nuclei that project to the cortex, such as the basal forebrain cholinergic nuclei (which produce acetylcholine), the locus ceruleus (which produces norepinephrine), the raphe nuclei (which produce serotonin), and certain nuclei of the thalamus.

Because the hippocampus and other medial temporal lobe structures are the earliest and most severely affected brain regions in Alzheimer's disease, episodic memory – and in particular the file cabinet component of episodic memory – is the earliest and most impaired cognitive function. Common symptoms include asking the same questions repeatedly, repeating the same stories, forgetting appointments, and leaving the stove on. Following Ribot's law, patients with Alzheimer's disease show *anterograde amnesia* or difficulty learning new information. They also show *retrograde amnesia* or difficulty retrieving previously learned information. However, the patients typically demonstrate preserved memory for remote information. Thus, a patient may report, "I've got short-term memory problems – I cannot remember what I did yesterday but I can still remember things from thirty years ago." Not understanding that this pattern is suggestive of the memory impairment common in Alzheimer's disease, family members may report that they feel confident that whatever the patient's problem is, that "it isn't Alzheimer's disease," because the patient can still remember what happened many years ago.

Another hallmark of the episodic memory impairment in the disease is that memory is impaired even when the learning or encoding of information is maximized by multiple rehearsals, and after retrieval demands have been minimized with the use of a multiple-choice recognition test. In other words, even when patients appear to have successfully learned new information by repeating it back over several learning trials, they are often unable to recognize this information on a multiple-choice test. This type of memory loss is often referred to as a "rapid rate of forgetting," although whether the information has been truly learned or not in the first place has been a matter of debate (e.g., Budson et al., 2007).

From a practical standpoint, this means that in very mild and mild Alzheimer's disease on the MMSE and the Blessed Dementia Scale, for example, the registration or learning of the items is usually intact, but recall is usually impaired (and although it is not part of these tests, patients are also unable to choose the registered words from a list). Similarly, on more comprehensive tests that include both recall and recognition components (such as the CERAD word list memory test (Welsh et al., 1992)), there will be a number of words successfully learned in the encoding trials, that are not recognized on the multiple-choice recognition test.

In addition to rapid forgetting, patients with Alzheimer's disease also experience distortions of memory and false memories. These distortions may include falsely remembering that they have already turned off the stove or taken their medications, leading patients to neglect performing these tasks. More dramatic distortions of memory may occur when patients substitute one person in a memory for another, combine two memories together, or think that an event that happened long ago occurred recently. Sometimes a false memory can be confused with a psychotic delusion or hallucination. For example, a patient may claim to have recently seen and spoken with a long-deceased family member. This patient is much more likely to be suffering from a memory distortion or a false memory than a true hallucination. The same is true for the patient who claims that people are breaking into the house, and moving around things. That these symptoms likely represent memory distortions rather than true hallucinations or delusions has implications when it comes time for treatment; as memory distortions are best treated with memory enhancing medications (such as cholinesterase inhibitors) rather than antipsychotic medications.

Although not as impaired as medial temporal lobes, the frontal lobes are involved in both Alzheimer's disease and mild cognitive impairment (Dickerson & Sperling, 2009; Schonknecht et al., 2009). Frontal lobe dysfunction may be one reason that patients with Alzheimer's disease experience memory distortions and false memories. That they show frontal lobe dysfunction also means that, in addition to a major impairment with their "file cabinet," patients with Alzheimer's disease also show milder but definite impairment with their "file clerk" as well. Additionally, patients with Alzheimer's disease have major pathology in parietal cortex, and this pathology may occur quite early in the disease (McKee et al., 2006). Thus, although the medial temporal lobe pathology is most relevant in the majority, patients with Alzheimer's disease have multiple components impaired in their episodic memory system.

Dementia with Lewy bodies

Because patients with dementia with Lewy bodies often have concomitant Alzheimer's disease pathology, the pattern of cognitive impairment may be identical to that which is seen in Alzheimer's disease, or it may show relatively greater impairment on measures of attention, visuospatial, and executive function with relative sparing of episodic memory. In general, however, episodic memory dysfunction in dementia with Lewy bodies is greater than that of other parkinsonian disorders such as multiple system atrophy and Parkinson's disease (Kao et al., 2009).

Vascular dementia

As vascular dementia is dementia due to multiple strokes, whether a patient experiences episodic memory dysfunction and of what quality depends upon the size, number, and location of the cerebrovascular disease. However, the most common type of vascular dementia is that due to small vessel ischemic disease (also known as subcortical ischemic vascular disease, see Chapter 2). Because small vessel ischemic disease has a predilection for the subcortical white matter, and because most of the white matter in the brain is carrying neural signals going to or from the frontal lobes, patients with vascular dementia experience a "frontal" memory disorder. In other words, because their frontal lobes are not working (or are not properly connected to the rest of the brain), they show impairment in the "file clerk" component of episodic memory. Consequently encoding is often impaired, as is free recall, whereas relative preservation is typically seen when tasks that assist in retrieval, such as cued recall and recognition, are used.

Frontotemporal dementia

Patient with frontotemporal dementia often show no impairment in episodic memory until late in the disease. If they do demonstrate early episodic memory impairment, it is generally that of a frontal memory disorder, similar to that of patient with vascular dementia.

Other disorders

Many other dementias also affect the frontal lobes and thus may also lead to a frontal episodic memory disorder. These dementias include progressive supranuclear palsy, corticobasal degeneration, multiple system atrophy, and normal pressure hydrocephalus.

Semantic Memory

Semantic memory refers to our store of conceptual and factual knowledge that is not related to any specific memory, such as the color of a banana or what a cup is used for. Like episodic memory, semantic memory is an explicit and declarative memory system. Evidence that semantic memory and episodic memory are separate memory systems has come from both neuroimaging studies (Schacter et al., 2000) and the fact that previously acquired semantic memory is spared in patients who have severe impairment of episodic memory, such as with disruption of Papez's circuit or surgical removal of the medial temporal lobes (Corkin, 1984).

In its broadest sense, semantic memory includes all our knowledge of the world not related to any specific episodic memory. It could therefore be argued that semantic memory resides in multiple cortical areas throughout the brain. For example, there is evidence that visual images are stored in nearby visual association areas (Vaidya et al., 2002). A more restrictive view of semantic memory justified in light of the naming and categorization tasks by which it is usually tested, however, localizes semantic memory to the inferolateral temporal lobes, particularly on the left (Figure 11.6, plate section) (Damasio et al., 1996; Perani et al., 1999).

Semantic memory dysfunction in dementia

The most common clinical disorder disrupting semantic memory is Alzheimer's disease (Salmon & Bondi, 2009). This disruption may be due to pathology in the inferolateral temporal lobes (Price & Morris, 1999) or to pathology in frontal cortex (Lidstrom et al., 1998), leading to poor activation and retrieval of semantic information (Balota et al., 1999). Supporting the idea that two separate memory systems are impaired in Alzheimer's disease, episodic and semantic memory decline independently of each other in this disorder (Green & Hodges, 1996).

Almost any disorder that can disrupt the inferolateral temporal lobes may cause impairment of semantic memory, including traumatic brain injury, stroke, surgical lesions, encephalitis, and tumors (Table 11.2). Patients with semantic dementia (the temporal variant of frontotemporal dementia) exhibit deficits in all functions of semantic memory, such as naming, single-word comprehension, and impaired general knowledge (such as the color of common items). Other aspects of cognition, however, are relatively preserved, including components of speech, perceptual and nonverbal problem-solving skills, and episodic memory (Hodges, 2001).

Although naming difficulties (particularly with proper nouns) are common in healthy older adults, naming difficulties may also be a sign of a disorder of semantic memory. When a disorder of semantic memory is suspected, the evaluation should include the same components as the work-up for episodic memory disorders. One of the first aspects of the history and cognitive examination that should be ascertained is whether the problem is solely one of difficulty in recalling people's names and other proper nouns (common in healthy older adults) or to a true loss of semantic information. Patients with mild dysfunction of semantic memory may show only reduced generation of words in a semantic category (for example, the number of grocery items that can be generated in one minute), whereas patients with a more severe impairment of semantic memory usually show a two-way naming deficit: they are unable to name an item when it is described, and they are also unable to describe an item when it is named. General knowledge is also impoverished in these more severely affected patients. Treatment will depend upon the specific disorder identified.

Procedural Memory

Procedural memory refers to the ability to learn cognitive and behavioral skills and algorithms that operate at an automatic, unconscious level. Procedural memory is non-declarative and implicit. Examples include learning to play the violin or drive a standard transmission automobile (Table 11.1). Because procedural memory is spared in patients who have severe deficits of the episodic memory system (such as those who have undergone surgical removal of the medial temporal lobes), it is clear that the procedural memory system is separate and distinct from the episodic memory system (Corkin, 1984; Heindel et al., 1989).

Functional imaging research has shown that a number of brain regions involved in procedural memory become active as a new task is learned, including the supplementary motor area, basal ganglia, and cerebellum (Daselaar et al., 2003) (Figure 11.6). Convergent evidence comes from studies of patients with damage to the basal ganglia or cerebellum who show impairment in learning procedural skills (Exner et al., 2002).

Procedural memory dysfunction in dementia

Because the basal ganglia and cerebellum are relatively spared in early Alzheimer's disease, despite their episodic memory deficit these patients show normal acquisition and maintenance of their procedural memory skills (Baird & Samson, 2009). Parkinson's disease is the most common disorder disrupting procedural memory (Wang et al., 2009), and thus it should not be surprising that the most common neurodegenerative disorder to disrupt procedural memory is dementia with Lewy bodies (Longworth et al., 2005). Patients in the early stages of Huntington's disease and olivopontocerebellar degeneration also show impaired procedural memory while performing nearly normally on episodic memory tests (Heindel et al., 1989; Salmon et al., 1998). Other causes of damage to the basal ganglia or cerebellum including tumors, strokes, and hemorrhages may also disrupt procedural memory. Patients with major depression also show impairment in procedural memory tasks, perhaps because depression involves dysfunction of the basal ganglia (Sabe et al., 1995).

Disruption of procedural memory should be suspected when patients show evidence of either substantial difficulties in learning new skills (compared to their baseline) or the loss of previously learned skills. For example, patients may lose the ability to perform automatic, skilled movements, such as knitting, swinging a golf club, or playing the saxophone. Although these patients may be able to relearn the fundamentals of these skills, explicit thinking becomes required for their performance. As a result, patients with damage to the procedural memory system lose the automatic, effortlessness of simple motor tasks that healthy individuals take for granted. The evaluation of disorders of procedural memory is similar to that of disorders of episodic memory; treatment depends upon the specific disease process. Lastly, it is worth noting that patients whose episodic memory has been devastated by a static disorder, such as encephalitis, have had successful rehabilitation by using procedural memory (and other non-declarative forms of memory) to learn new skills (Glisky & Schacter, 1989).

Working Memory

Bringing together the traditional fields of attention, concentration, and short-term memory, working memory refers to the ability to temporarily maintain and manipulate information that one needs to keep in mind. Requiring active and conscious participation, working memory is an explicit and declarative memory system. Working memory has traditionally been divided into three components: one that processes phonologic information (e.g., keeping a phone number "in your head"); one that processes spatial information (e.g., mentally following a route); and an executive system that allocates attentional resources (Baddeley, 1998).

Studies have demonstrated that working memory involves a network of cortical and subcortical areas, which differ depending on the particular task (Rowe et al., 2000). Participation of the prefrontal cortex, however, is involved in virtually all working memory tasks (Fletcher & Henson, 2001) (Figure 11.6). The network of cortical and subcortical areas typically includes posterior brain regions (e.g., visual association areas) that are linked with prefrontal regions to form a circuit. Research suggests that spatial working memory tends to involve more regions on the right side, and phonologic working memory tends to involve more regions on the left side of the brain. Bilateral brain activation is observed, however, in more difficult working memory tasks, regardless of the nature of the material being processed (Newman et al., 2003). Additionally, an increase in the number of brain regions activated in prefrontal cortex is observed as the complexity of the task increases (Jaeggi et al., 2003).

Working memory dysfunction in dementia

Because working memory depends upon networks which include frontal and parietal cortical regions as well as subcortical structures, most neurodegenerative diseases impair working memory. Studies have demonstrated that working memory may be impaired in patients with Alzheimer's disease, Parkinson's disease, Huntington's disease, dementia with Lewy bodies, as well as less common disorders such as progressive supranuclear palsy (Table 11.2) (Calderon et al., 2001; Gotham et al., 1988). In Alzheimer's disease spatial working memory is particularly impaired (Iachini et al., 2009). In addition to these neurodegenerative diseases, almost any disease process that disrupts the frontal lobes or their connections to posterior cortical regions and subcortical structures can interfere with working memory. Such processes include tumors, strokes, multiple sclerosis, head injury, and others (Kubat-Silman et al., 2002; Sfagos et al., 2003). Because it involves the silent rehearsal of verbal information, almost any type of aphasia may impair phonologic working memory. Disorders that diminish attentional resources, including attention deficit hyperactivity disorder, obsessive compulsive disorder, depression, and schizophrenia, can also impair working memory (Egeland et al., 2003; Klingberg et al., 2002; Purcell et al., 1998).

Disorders of working memory may present in several different ways. Often the patient will exhibit an inability to concentrate or pay attention. Impairment in performing a new task with multi-step instructions is frequently seen. Interestingly, a disorder of working memory may also present as a problem with episodic memory, because information must first be "kept in mind" by working memory in order for episodic memory to encode it (Fletcher & Henson, 2001). Such cases will therefore show a primary impairment in encoding.

The evaluation of disorders of working memory is similar to that of disorders of episodic memory. Treatment depends upon the underlying cause. Stimulants, approved by the FDA for the treatment of attention deficit hyperactivity disorder (Elia et al., 1999; Mehta et al., 2004), will often be helpful in disorders of working memory.

Conclusion

Although traditionally memory has been viewed as a simple concept, converging and complementary evidence from patient studies and more recent neuroimaging research suggest that memory is composed of separate and distinct systems. Improved understanding of these different types of memory will aid the clinician in the diagnosis and treatment of dementia and other causes of memory impairment in their patients. As more specific therapeutic strategies are developed for the treatment of diseases which cause memory dysfunction, this knowledge will become increasingly important.

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